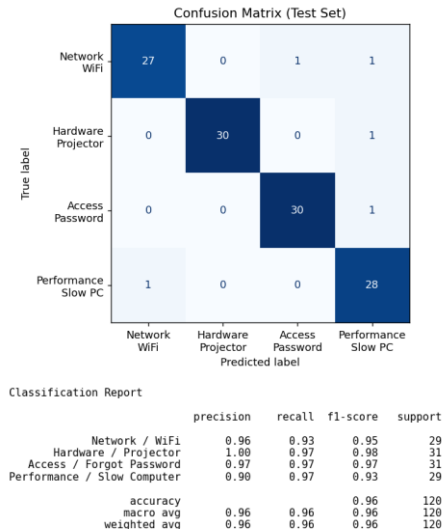


Project Development Phase Model Performance Test

Date	02 October 2026
Team ID	SWTID-2026-8568
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	Classification Model (Multinomial text classification – TF-IDF + Logistic Regression): Dataset: 600 tickets (Train 480 / Test 120); 4 classes – Network/WiFi, Hardware/Projector, Access/Forgot Password, Performance/Slow Computer. Confusion Matrix - rows = actual, columns = predicted (order: WiFi, Projector, Password, Slow PC) <pre>[27 0 1 1] [0 30 0 1] [0 0 30 1] [1 0 0 28]</pre> Accuracy Score - 95.83% (0.9583) Classification Report - Macro avg Precision 0.96, Recall 0.96, F1-score 0.96; per-class F1: WiFi 0.95, Projector 0.98, Password 0.97, Slow PC 0.93. Regression Model: Not applicable (problem is classification).	

2.	Tune the Model	Hyperparameter Tuning - GridSearchCV over TF-IDF ngram_range {(1,1),(1,2)} and Logistic Regression C {0.1, 1, 10, 100}. Best: ngram_range=(1,2), C=1. Validation Method - 80/20 stratified hold-out split plus 5-fold Stratified Cross-Validation on the training set (best CV accuracy 96.67%). Baseline test accuracy 95.83% vs tuned 95.83%.	<pre>GridSearchCV (5-fold Stratified CV) Parameter grid: tfidf_ngram_range : (1,1), (1,2) clf__C : 0.1, 1, 10, 100 Best parameters: ngram_range = (1, 2) C = 1 Best CV accuracy : 0.9667 Baseline test acc: 0.9583 Tuned test acc : 0.9583</pre>
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